



Fig. 6: Initial information after switch 'on'



Fig. 7: Current value of measured temperature



Fig. 8: Recorded minimum and maximum temperature values

Once the pulse width (pulse duration = t_{out}) of the input signal is known, the frequency of the signal can easily be computed as:

$$f_{\text{out}} = 1/t_{\text{out}}$$

The temperature can be computed as $\text{CPU_FREQ}/(2 \times \text{width})$.

Software

This integration and functionality of the temperature indicator are achieved by the software program. The software was developed in Win AVR environment, which is available on the Internet free of cost for both Windows and Linux operating systems. Win AVR is capable of handling all the AVR functionalities like UART, timer, ADC and interrupts, and offers the facility of writing the program in both Assembly and 'C' languages.

The software for this project was written in 'C' language and Win AVR was used to compile the 'C' source program for debugging purpose. Fi-

nally, the .hex file was downloaded to the microcontroller using a suitable programmer.

Functioning of the LCD and switch, as also temperature calculation, are achieved through the software control. For better control over the software, the software has been written in modular form, i.e., the related functions are put in specific files. This makes the software much easier to code, control and debug.

The source program files used in this project are shown in Table II.

The pwm.c file contains the code to integrate, compute and control the LCD, push-to-on switch and data received from MAX6577. The main sub-routines used in this file code are:

1. `get_Sensor_Data()`: As described earlier, this code is used to detect the first leading edge of the temperature sensor's output signal (and start the internal timer of the microcontroller), and then detect the first trailing edge of the signal (to stop the timer). Subsequently, the timer counts are used to compute the pulse-width and frequency.

2. `width_Temp_Convert()`: The code in this subroutine is used to convert the measured pulse-width data into equivalent temperature values (including minimum and maximum recorded values) and display the results at appropriate locations depending on the information desired.

In while(1) loop of the operation, the microcontroller checks whether the push-to-'on' switch has been pressed or not and takes action accordingly (displays the minimum and maximum temperature values). In this loop operation, the software also checks whether the latest temperature reading is lower/higher than the previously recorded value and sets the minimum/maximum temperature variables. It also helps in displaying the desired information accordingly.

The lcd.c file contains the code to control the functioning of the LCD module. The code controls the initialization of the LCD, data writing on the

LCD, and also the movement, characteristics and location of the cursor. It offers the facility to write data on the LCD character-by-character or string-wise. The command set used in the software is based on the command set used in the LCD module.

The lcd.h header file contains all the constant variable values and names of the sub-routines used by the various files in the project. It clearly indicates which variable can be used as a global variable and which of the sub-routines can be accessed and used across the various software files used in the project.

The key.c code file is used to check the status of the push-to-'on' switch (whether it has been pressed or not). The pushbutton is used to display the recorded minimum and maximum temperature values at appropriate locations and then returned to its original point of operation.

Operation

When the circuit is switched on, the information is displayed on the LCD module as shown in Fig. 6. Thereafter, the circuit shows the latest measured temperature value (see Fig. 7).

In the background, both minimum and maximum temperature values are recorded. The same can be viewed by pressing switch S2 (refer Fig. 8). The maximum temperature value is shown by letter 'H' and the minimum temperature value is shown by letter 'L'.

Now the unit continues to display the monitored, minimum and maximum temperature values simultaneously until switch S2 is pressed again. Pressing switch S2 returns the unit to its initial point of operation (see Fig. 7).

The source code can be used as it is or modified but the concept remains the same for reading any PWM data (like gravity and voltage) from the sensor.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD. ●